

5th Japan Astronomy Olympiad

Practical Examination — Answer Sheet Problem 1

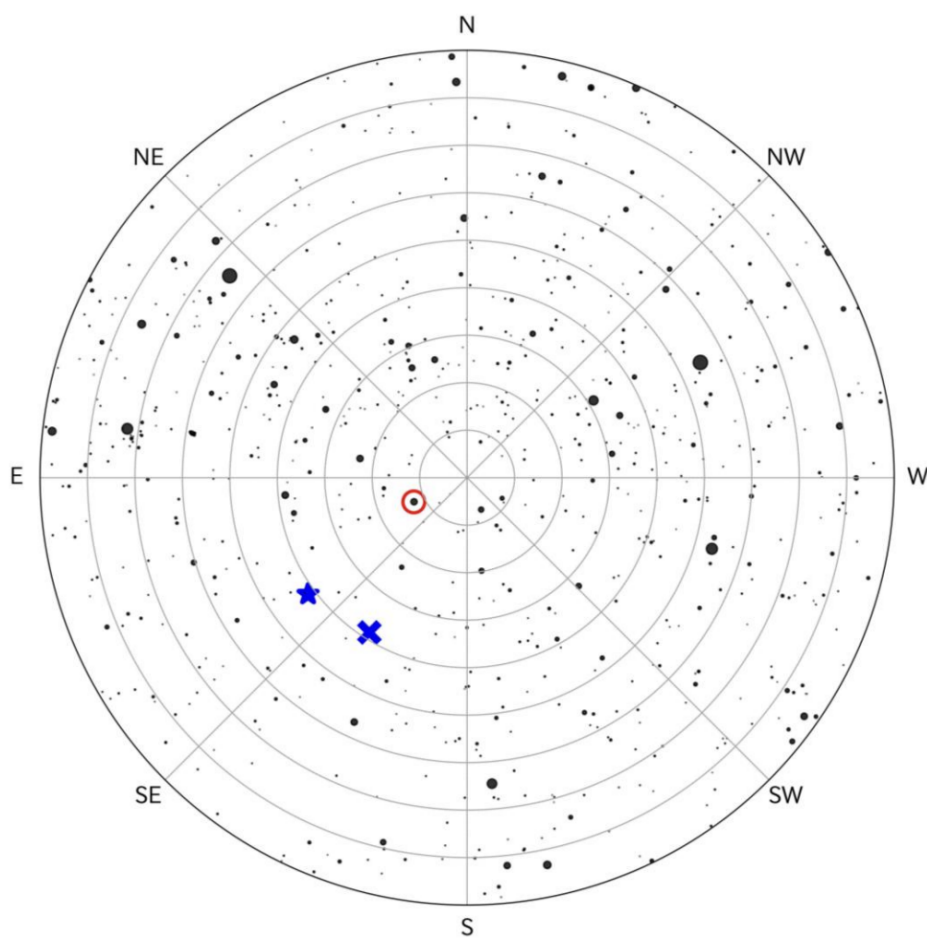
Question 1

(1)	Proper name:	Bayer designation:
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(2) *Questions 1(2) through Question 3(1) — all three — should be answered on the same star chart below.*

Question 2

Question 3 (1)



(2) Answer (minimum field of view):

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Examinee No.: _____ Name: _____

Question 4

(1) Choice (circle one): **Mars** **Saturn** **Uranus** **Neptune**

Reasoning / calculation:

Answer (orbital period): _____

(2) *Reasoning / calculation:*

Answer (altitude at culmination): _____

(3) *Reasoning / calculation:*

Answer (local sidereal time at culmination): _____

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Practical Examination — Answer Sheet Problem 2

Question 1

Reasoning / calculation:

Answer ($S/N \propto t^?$): _____ Answer ($S/N \propto w^?$): _____

Question 2

(1) *Reasoning / calculation:*

Answer (exposure time): _____

(2) *Reasoning / calculation:*

Answer (exposure time): _____

Question 3

(1) *Reasoning / calculation:*

Answer: _____

(2) Answer / explanation:

5th Japan Astronomy Olympiad Practical Exam — Problem 2
(continued)

Examinee No.: _____ Name: _____

Question 4 Answer / explanation:

Question 5 Answer / explanation:

Question 6

Corrected values = (Light – Dark)/Flat. Fill in the grid below.

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Practical Examination — Answer Sheet Problem 3 (Sheet 1)

Question 1

(1) (A): _____ (B): _____ (C): _____ (D): _____

(2) *Reasoning / calculation:*

Answer (Lyman- α wavelength): _____ Answer (Lyman limit wavelength): _____

(3) *Reasoning / calculation:*

Answer (redshift): _____

(4) Choice (circle one): ① $z = 8$ ② $z = 10$ ③ $z = 12$ ④ $z = 14$

Reasoning:

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(Sheet 1, continued)

Examinee No.: _____ Name: _____

Question 2

(1) *Reasoning / calculation:*

Answer (survey volume): _____

(2) Fill in the table below.

M_{UV} bin	$\Phi(M_{\text{UV}})$ [$\text{Mpc}^{-3} \text{mag}^{-1}$]
$-23.0 \sim -22.5$	
$-22.5 \sim -22.0$	
$-22.0 \sim -21.5$	
$-21.5 \sim -21.0$	
$-21.0 \sim -20.5$	
$-20.5 \sim -20.0$	
$-20.0 \sim -19.5$	
$-19.5 \sim -19.0$	

Note: Question 2(3) should be answered on answer graph sheet ①.

(4) *Derivation:*

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Practical Examination — Answer Sheet Problem 3 (Sheet 2)

Question 2 (continued)

(5) *Reasoning / calculation:*

Answer (ρ_{UV}): _____

(6) Fill in the SFR density table below.

z bin	SFR density [$M_{\odot} \text{ yr}^{-1} \text{ Mpc}^{-3}$]
0.1 $\sim$ 0.3	
0.3 $\sim$ 0.5	
0.5 $\sim$ 0.7	
0.7 $\sim$ 1.1	
1.1 $\sim$ 1.5	
1.5 $\sim$ 1.9	
1.9 $\sim$ 2.3	
2.3 $\sim$ 2.7	
2.7 $\sim$ 3.1	
3.1 $\sim$ 3.5	
3.5 $\sim$ 4.5	
4.5 $\sim$ 5.5	
5.5 $\sim$ 6.5	
6.5 $\sim$ 7.5	
7.5 $\sim$ 8.5	

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(Sheet 2, continued)

Examinee No.: _____ Name: _____

Question 2 (continued)

Note: Question 2(7) graph should be answered on answer graph sheet ②.

(7) Redshift at which SFR density peaks (to 1 significant figure):